

HW3:

First, we must define the appropriate membership functions for each input variable. Then, we perform fuzzification, mapping each crisp input value to its corresponding degree of membership in the fuzzy sets.

Input (Error range: $[-5, +5]$):

- Cold: Between -5 and 0
- Normal: Between -2 and $+3$
- Hot: Between $+2$ and $+5$

Output (Heater Power range: $[0, 100]$):

- Low: Between 0 and 40
- Normal: Between 30 and 70
- High: Between 60 and 100

Rules:

1. If cold then high
2. If medium then medium
3. If hot then low

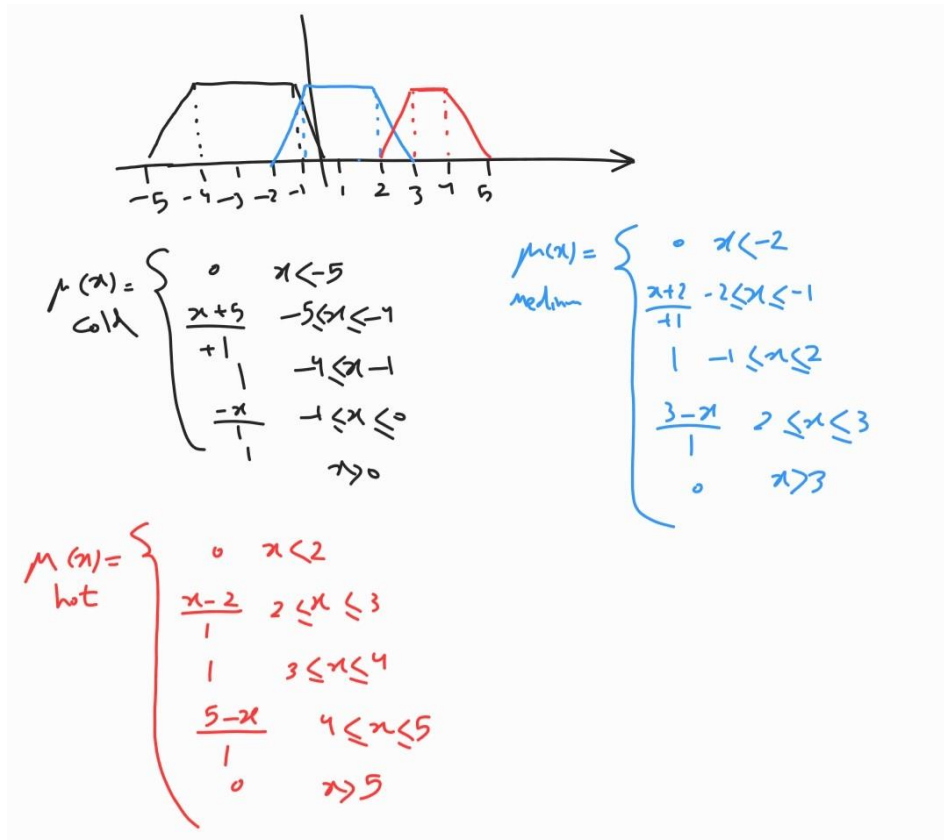
For all three inputs and outputs, we use trapezoidal membership functions:

$$\mu(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c \leq x \leq d \\ 0, & x > d \end{cases}$$

Then we calculate the centroid using this formula:

$$S_{\text{crisp}} = \frac{\int S \cdot \mu_S(S) dS}{\int \mu_S(S) dS}$$

input membership functions:



output membership functions:

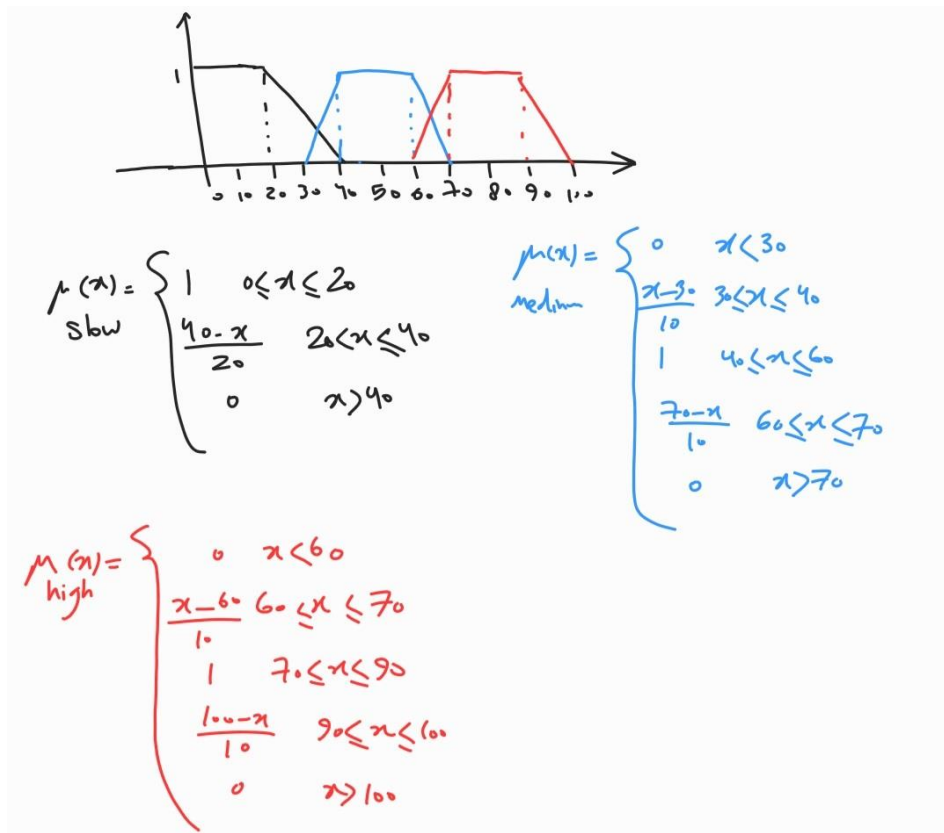


Diagram of input membership functions:

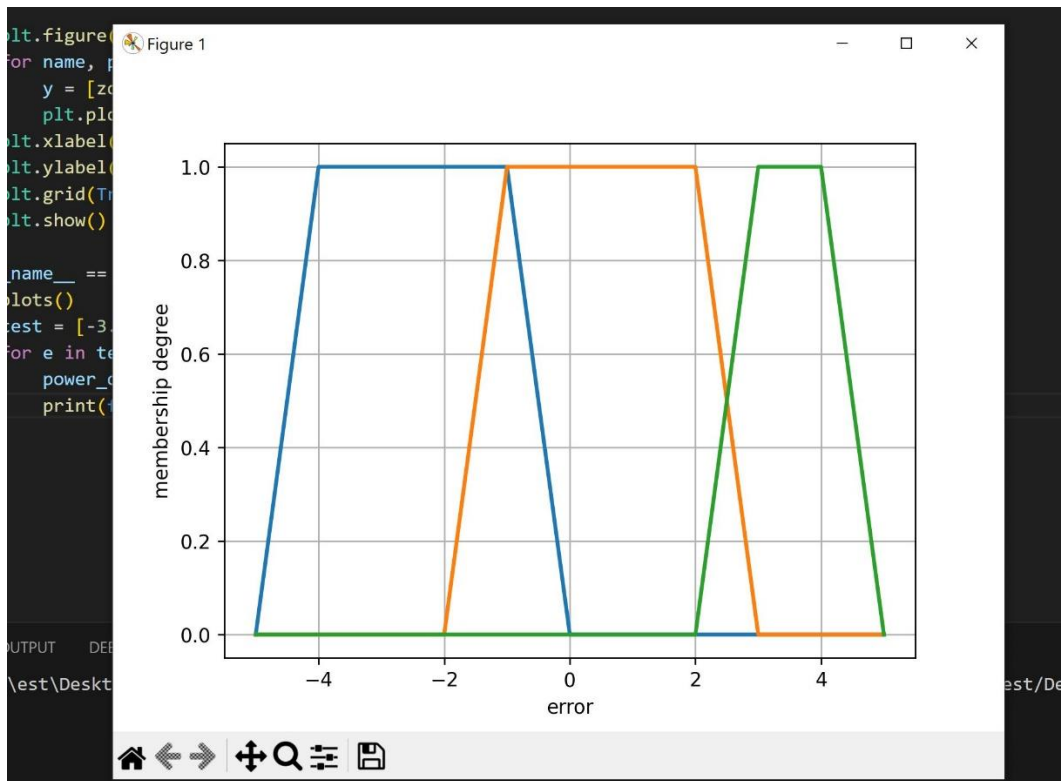
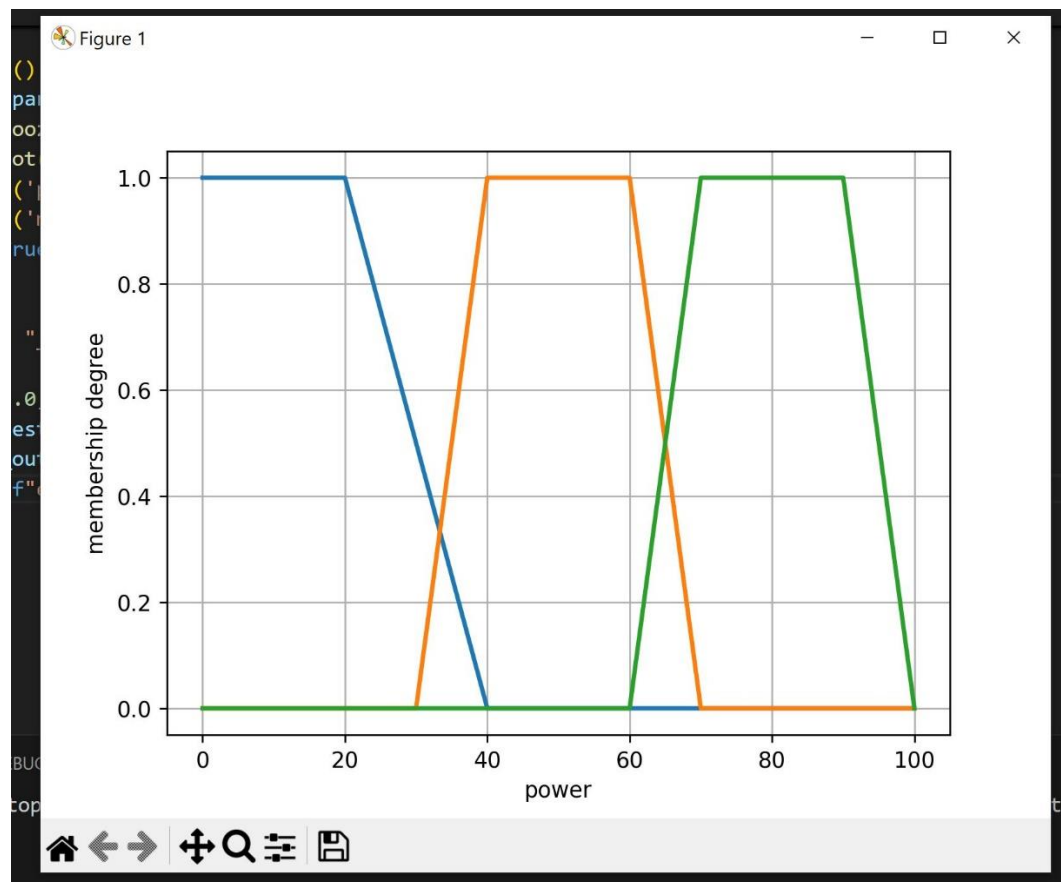


Diagram of output membership functions:



Output:

```
PS C:\users\est\Desktop\hnmw> & C:/users
error = -3.0°C -> power = 80.0%
error = -1.0°C -> power = 65.0%
```

Output explanation:

3- به تابع $load$ را با مقدار 1 قطع می‌کنند ← طبق قانون شماره 1، تابع $load$ را 100
 فعال می‌کنند ← محور x در $load$ نشان داده شده که $load$ تابع $high$ باشد که 100 شود
 پس centroid این بخش صحیح شود :

$$Num: \int_{60}^{70} s \cdot \frac{(s-60)}{10} ds + \int_{70}^{90} s \cdot ds + \int_{90}^{100} s \cdot \left(\frac{100-s}{10}\right) ds = 2400$$

$$Den: \int_{60}^{70} \frac{(s-60)}{10} ds + \int_{70}^{90} ds + \int_{90}^{100} \frac{100-s}{10} ds = 30 \quad \Rightarrow S = \frac{Num}{Den} = 80$$

1- به تابع $load$ را در $medium$ قطع می‌کنند. $load \leftarrow cold$
 $load \leftarrow medium$
 طبق قوانین 1 و 2، تابع $high$ ، $medium$ فعال می‌شوند
 توجه این توابع $\begin{cases} y=1 \leftarrow high \\ y=1 \leftarrow medium \end{cases}$

$$Den: \int_{30}^{40} \frac{s-30}{10} ds + \int_{40}^{60} ds + \int_{60}^{65} \frac{70-s}{10} ds + \int_{65}^{70} \frac{s-60}{10} ds + \int_{70}^{75} ds + \int_{75}^{100} \frac{100-s}{10} ds$$

$$= 57.5$$

$$Num: \int_{30}^{40} s \cdot \frac{(s-30)}{10} ds + \int_{40}^{60} s \cdot ds + \int_{60}^{65} s \cdot \left(\frac{70-s}{10}\right) ds + \int_{65}^{70} s \cdot \left(\frac{s-60}{10}\right) ds + \int_{70}^{75} s \cdot ds +$$

$$\int_{75}^{100} s \cdot \frac{(100-s)}{10} ds = 466$$

$$\rightarrow S = \frac{Num}{Den} = 65$$